

Department of Information and Communication Technology

The Lake Hotel & Resort – Hotel Management System

Project Proposal
Group Project 2025
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Abstract

This project proposes the development of a Hotel Management System (HMS) for G.H.S. Sudesh, a hotel currently relying on manual operations for room accommodation, event hosting, pool access, buffet dining, and reporting. The manual system is time-consuming, error-prone, and inefficient, leading to operational challenges and customer dissatisfaction.

The proposed HMS will automate and streamline all hotel operations, including room bookings, event management, pool access control, buffet dining reservations, and real-time reporting. The system will be developed using modern technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js and MongoDB, ensuring scalability, user-friendliness, and security. By implementing this system, the hotel aims to enhance operational efficiency, improve customer experience, and generate accurate reports for better decision-making.

This project will be accomplished through a structured development process, including requirement analysis, system design, implementation, testing, and deployment. The final product will be a web-based application accessible to both hotel staff and customers.

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1. Introduction and Description of the Project

The hospitality industry is one of the most dynamic and competitive sectors, where customer satisfaction and operational efficiency are critical to success. In today's digital age, hotels are increasingly adopting technology to streamline their operations, enhance guest experiences, and stay ahead of the competition. However, many small to medium-sized hotels still rely on manual systems for managing their day-to-day operations, leading to inefficiencies, errors, and missed opportunities for growth.

This project focuses on developing a Hotel Management System (HMS) for G.H.S. Sudesh, a hotel that currently uses a manual system for all its operations, including room accommodation, event hosting, pool access, buffet dining, and reporting. The manual system is not only time-consuming but also prone to errors, making it difficult for the hotel to manage its operations effectively and provide a seamless experience to its guests.

The proposed HMS will serve as a comprehensive, automated solution to address these challenges. It will integrate all aspects of hotel operations into a single, centralized platform, enabling the hotel to manage room bookings, event registrations, pool access, buffet dining reservations, and reporting with ease. By automating these processes, the system will significantly reduce manual effort, minimize errors, and improve overall efficiency.

The system will also enhance the guest experience by providing a user-friendly interface for online bookings, real-time updates, and seamless service delivery. Additionally, the reporting module will offer valuable insights into the hotel's performance, enabling data-driven decision-making and strategic planning.

This project is not just about developing a software solution; it is about transforming the way the hotel operates and interacts with its guests. By leveraging modern technologies and best practices in software development, we aim to deliver a system that is scalable, secure, and easy to use. The successful implementation of this system will not only improve the hotel's operational efficiency but also position it as a modern, customer-centric establishment in the competitive hospitality market.

1.1 Client Details

Client Name: G.H.S. Sudesh

Email: shanakaqs@gmail.com

Contact Details: +94771389482

Business Need: The client requires a comprehensive software solution to automate their manual operations, including room accommodation, event hosting, pool access, buffet dining, and reporting. The current manual system is inefficient and prone to errors, leading to operational delays and customer dissatisfaction.

1.2 Problem Specification

The hotel currently uses a manual system for all operations, which involves.

- **Manual room booking and tracking-** The hotel currently uses Google Forms and paper-based records to manage room bookings, which is time-consuming and prone to errors, making it difficult to track room availability accurately.
- **Handwritten event registrations for weddings, birthdays, meetings, and family outings-** Event registrations are done manually, leading to inefficiencies, potential booking overlaps, and difficulties in managing event details and payments.
- **Physical tickets for pool access-** Pool access is managed using physical tickets, which limits real-time monitoring of pool capacity and creates inconvenience for guests and staff.
- **Manual buffet dining reservations-** Buffet dining reservations are handled manually, causing delays, errors in booking confirmations, and challenges in managing dining capacity.
- **Time-consuming and error-prone reporting processes-** Generating reports on room occupancy, events, and other metrics is done manually, resulting in delays, inaccuracies, and inefficiencies in decision-making.
- **Some operations are done by using Google Forms-** While Google Forms are used for certain tasks, they lack integration with other systems, leading to fragmented data and inefficiencies in managing operations.

Challenges Faced:

- Inefficient and time-consuming operations- Manual processes slow down operations, leading to delays in service delivery and increased workload for staff.
- Elevated risk of human errors in bookings and reporting- Handwritten records and manual calculations often result in mistakes, causing booking conflicts and inaccurate reports.
- Difficulty in managing multiple events and services simultaneously- Handwritten records and manual calculations often result in mistakes, causing booking conflicts and inaccurate reports.
- Lack of real-time data for decision-making- Without real-time updates, the hotel management cannot access accurate data, hindering timely and informed decision-making.
- Poor customers experience due to delays and errors- Guests face inconveniences such as booking errors, long wait times, and service delays, leading to dissatisfaction and a negative impression of the hotel.

1.3 Solution Outline

The proposed Hotel Management System (HMS) will address these challenges by providing a centralized, automated platform for managing all hotel operations. The system will include the following modules:

- Room Accommodation Management: Online room booking, check-in/check-out, and availability tracking.
- Event Hosting Management: Event registration, scheduling.
- Pool Access Management: Digital pool access tickets and real-time capacity monitoring.
- Buffet Dining Management: Online buffet reservations.
- Reporting Module: Real-time reports on room occupancy, event bookings, pool usage, and dining reservations.

1.4 Key Benefits

Improved Efficiency: Automation of manual processes will save time and reduce errors.

Enhanced Customer Experience: Faster bookings, real-time updates, and seamless service delivery.

Accurate Reporting: Real-time data for better decision-making and performance analysis.

- **Scalability:** The system can be expanded to include additional features in the future.

Cost Savings: Reduced operational costs through automation and error reduction.

2.Aim and Objectives

Aim:

To design and develop a comprehensive Hotel Management System that automates and streamlines all hotel operations, enhancing efficiency, customer satisfaction, and decision-making capabilities.

Objectives:

- **Develop a User-Friendly Interface:** Design and implement an intuitive and responsive user interface for both hotel staff and customers, ensuring ease of navigation and a seamless user experience.
- **Implement Room Booking and Management:** Develop a module for online room booking, check-in/check-out, and real-time room availability tracking to streamline the reservation process.
- **Event Hosting Management:** Create a system for event registration, scheduling, and management, allowing the hotel to handle weddings, meetings, and other events efficiently.
- **Pool Access Management:** Integrate a digital pool access system with real-time capacity monitoring to enhance guest convenience and improve pool management.
- **Buffet Dining Reservations:** Develop an online buffet reservation system to automate dining bookings and manage dining capacity effectively.
- **Real-Time Reporting and Analytics:** Implement a reporting module that provides real-time updates on room occupancy, event bookings, pool usage, and dining reservations, enabling data-driven decision-making.
- **Ensure Data Security:** Implement robust security measures, including encrypted data storage and access controls, to protect sensitive guest and operational data from unauthorized access or breaches.
- **Automate Inventory and Resource Management:** Design automated processes to manage hotel resources (e.g., room inventory, pool capacity, dining reservations) based on real-time updates, reducing manual effort and minimizing errors.
- **Provide Comprehensive Reporting:** Develop reporting features that allow hotel management to generate customizable reports and analyze key performance metrics, helping them make informed decisions and identify trends.
- **Ensure Compatibility and Scalability:** Ensure that the web application is compatible with various devices and browsers, and design it with scalability in mind to accommodate future growth and additional features.
- **Provide Training and Support:** Offer comprehensive training sessions and ongoing support to hotel staff to ensure smooth adoption and effective utilization of the new system.

- Documentation: Prepare comprehensive documentation, including user manuals and technical guides.

3.Procedures.

3.1 Flow of the Project (15-Week Schedule)

This project will utilize the Scrum framework, emphasizing iterative development, continuous feedback, and adaptability. The project is estimated to span 15 weeks, targeting completion around May 19, 2025. We will use two-week sprints.

Sprint 1: Project Initiation

Activities-

- Visit The Lake Hotel & Resort to understand their current processes and needs.
- Conduct interviews and workshops with Mr. Sudesh and hotel staff
- Analyze gathered information and document detailed requirements for each feature.
- Prepare Project proposal
- Send Project proposal to client
- Group Meeting

Milestone- Fulfill the requirements as per client feedback

Sprint 2: Planning and Design

Activities-

- Finalize the project proposal report
- Create wireframes for the user interface (UI).
- Group Meeting
- Starting to Prepare System Requirements Specification Documentation

Milestone: Develop a comprehensive idea about the work system.

Sprint 3: System Design

Activities-

- Design the Entity-Relationship (ER) Diagram
- Group Meeting
- Design the UML diagrams, including use-case diagrams, activity diagrams, and data flow diagrams (DFD)
- Finalize the SRS Document

Milestone: Complete the system designing phase

Sprint 4: Implementation of the Core Functionality

Activities-

- Set up a GitHub repository for version control.
- Create MongoDB or MYSQL Database for ER diagram
- Setup the Project Environment:
 - o Setup React.js for Backend Development
 - o Setup Express.js, Node.js for Frontend Development
- Develop the Web Sites for the user visibility
- Conduct unit testing to ensure functionality

Milestone: Achieve a prototype of the project.

Sprint 5: Admin Functionality

Activities-

- Develop the admin (Manager) panel functionality, including:
 - o Admin's profile management
 - o View all reservations of customer's
 - o Approve booking user's reservation request.
 - o View current room, pool availability.

- o Manually booking events
- o manage Receptions admin (CRUD).
- o Manage the room, pool, events package
- Conduct unit testing to ensure functionality.

Milestone: Complete the admin (Manager) part of the application.

Sprint 6: Sub Admins, Customers functionality

Activities-

- Develop the Sub Admins (Reception) functionality, including:
 - o Food order (dining or trackway)
 - o Profile Management
 - o Room Booking
 - o pool booking
 - o View Customer Reservation
- Conduct unit testing to ensure functionality
- Develop the Customers functionality, including:
 - o Customer Registration
 - o Profile Management
 - o Room booking
 - o Events (weddings/Birthday parties/meetings) booking
 - o pool booking
 - o View and change self-reservation
 - o Feed back
- Conduct unit testing to ensure functionality

Milestone: Complete the Sub Admins (Reception) and Customers part of the application.

Sprint 7: Testing and Bug Fixing

Activities-

- Perform integration testing to ensure all parts work together smoothly.
- Conduct system testing to validate the entire system's functionality
- Address any bugs or issues identified during testing.
- Debugging and finish the product

Milestone: Deliver a fully functional product without any bugs

3.2 Project Plan

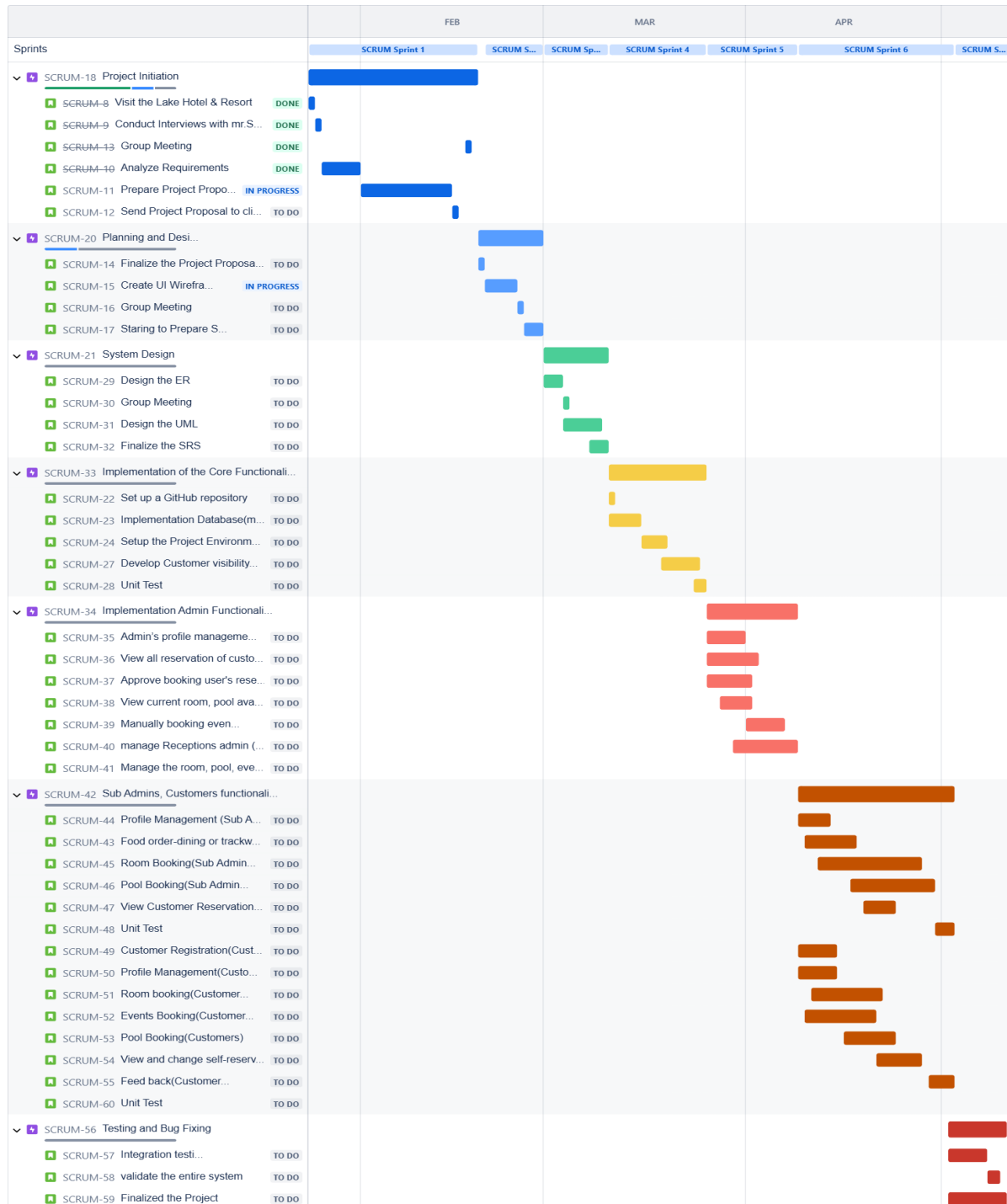


Figure 1: Grant Chart

4.Team Members And Roles

	Name	Function	Brief Description of the Function
1	TG/2021/1013- R.A.H.M.Dissanayaka	• User Profile • User Dashboard • Room Booking (user) • Pool Booking(user) • View and Changes self-Reservation (User) • View Room Availability (Admin) • View Pool Availability (Admin)	• Allows users to manage their personal information. • Providing an overview of users accounts, upcoming reservations, recent activity, and quick access to other features. • Enables users to reserve rooms with or without packages for specific dates and durations. • Allows users to reserve time slots with or without package. • This function allows users to check the status of their existing reservations (rooms or pool), including confirmation details, dates, times, and any other relevant information. It likely allows them to modify or cancel reservations within certain limits. • A feature for administrators to see the current availability of all rooms, likely presented in a calendar or grid format, allowing them to manage bookings and allocate resources efficiently.

			• This feature allows administrators to view the availability of pool facilities (e.g., time slots, cabanas), enabling them to manage bookings and prevent overbooking.
2	TG/2021/1016-L.S.Lakshan	•User registration(user) •User login(user) •Admin Dashboard •View all reservation(admin) •Approve Booking(Admin) •Events booking (User)	• Allows new users to create accounts by providing necessary information (e.g., name, email, password) to access the system's features. This typically involves verifying the user's email address. • Enables registered users to access their accounts by entering their credentials (usually username/email and password). • A centralized interface for administrators to manage the system, providing an overview of key metrics, recent activity, and quick access to administrative functions. • A feature for administrators to see a comprehensive list of all current and past reservations, including user details, booking dates, event details, and status. • Allows administrators to review pending reservations and manually approve or reject them. This might

			be necessary for certain types of bookings or to manage capacity. • Enables users to browse and book available events, selecting the desired event, date, time, and number of attendees. This might include features for purchasing tickets or making reservations.
3	TG/2021/1040- H.G.J.U. Gunathunga	• Admin login • Manually Booking Event • Menu Management • Manage Receptionist	• Develop a secure admin login system to allow authorized personnel to access the admin dashboard. • Implement a feature for admins to manually book events (e.g., weddings, meetings) on behalf of guests. • Create a module for admins to manage the buffet menu, including adding, updating, and deleting items. • Develop functionality for admins to add, update, or remove receptionist accounts and assign roles.
4	TG/2021/1068 - I.M.T.C.N.Bandara	• Receptionist login(receptionist) • Receptionist Profile • Receptionist Dashboard	• Enables authorized receptionists to access the system using their credentials (username/email and password). • Allows receptionists to manage their personal information, such as

		•Room booking (Receptionist) •Pool booking (Receptionist) •View Customer reservation(receptionist)	contact details and potentially other relevant information. • A dedicated interface for receptionists providing an overview of daily tasks, upcoming bookings, and quick access to frequently used functions. • Allows receptionists to book rooms on behalf of guests, either in person or over the phone. This likely includes selecting room type, dates, and handling guest information. • Enables receptionists to book pool facilities (e.g., cabanas, time slots) for guests, similar to room booking. • Allows receptionists to view, and potentially manage, existing customer reservations (rooms and pool), enabling them to assist guests with inquiries, changes, or cancellations.
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5. Hardware and Software Requirements

Development Environment

1. Hardware Requirements:

Laptops/Desktops: Development team members will require modern laptops or desktops with the following minimum specifications:

- Processor: Intel i5 or equivalent.
- RAM: 8GB.
- Storage: 256GB SSD.
- Operating System: Windows 10/11, macOS, or Linux.
- Internet Connection: A stable high-speed internet connection is required for collaboration, accessing cloud services, and downloading dependencies.

2. Software Requirements:

- Development Stack: The application will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Code Editor: Visual Studio Code (recommended) or any preferred IDE.
- Version Control: Git and GitHub/GitLab for source code management.
- Database: MongoDB (cloud-based or local installation).
- Runtime Environment: Node.js and npm/yarn for backend and frontend dependencies.
- Browser: Modern browsers like Google Chrome, Mozilla Firefox, or Microsoft Edge for testing.
- Additional Tools: Postman (for API testing), Docker (optional for containerization), and Azure CLI (for Azure cloud integration).

Deployment Environment

1.Cloud Platform: The application will be hosted on Microsoft Azure. The following Azure services will be used:

- Azure App Service: For hosting the web application.
- Azure Cosmos DB or Azure Database for MongoDB: For database storage.
- Azure Blob Storage: For storing static files (e.g., images, documents).
- Azure DevOps: For CI/CD pipeline setup (optional but recommended).

2.Server Specifications:

- App Service Plan: Basic or Standard tier (depending on expected traffic).
- Database Tier: Scalable tier based on data storage and performance requirements.
- Bandwidth: Sufficient bandwidth to handle user traffic and data transfer.

Client Expectations

1.Hardware: The client does not need to provide any hardware for development or deployment. However, they should ensure their team has access to modern devices (laptops/desktops) and a stable internet connection for accessing the application.

2.Software: The client should ensure their team uses modern browsers (e.g., Chrome, Firefox, Edge) to access the application. No additional software installations are required on the client side.

3.Support: The client may need to provide access to their Azure account for deployment and integration purposes.

6. Budget

We have estimated the cost of the project and justified it by the following cost factors.

Category	Item	Cost
Cloud Services	Web Hosting (VPS)	Rs. 15,000.00
Domain Registration	.com domain	Rs. 3,600.00
Equipment	Computer	Rs. 50,000.00
Total		Rs. 68,600.00

7.Risk Assessment

Challenges Faced:

- Lack of Project Management
 - Initially, the team faced challenges in project management, resulting in difficulties in task prioritization and time management.
- Lack of Framework Knowledge (React.js, Node.js)
 - The team encountered difficulties due to limited knowledge of the React.js and Node.js frameworks, which are essential for developing the frontend and backend of the application.
- Managing Team Members
 - With a diverse team, including members with varying levels of experience, coordinating tasks and ensuring effective communication was a challenge.
- Tight Deadlines
 - The project had strict deadlines, requiring efficient planning and execution to deliver the software on time.
- Working with APIs
 - Integrating APIs for data communication between the frontend and backend posed a challenge due to the complexity of the process

Overcoming Challenges

- Project Management
 - The team adopted the ClickUp project management tool to organize tasks, set priorities, and track progress. Additionally, team members applied project management principles from their academic studies to improve time and resource management.
- Framework Knowledge
 - To improve their understanding of React.js and Node.js, team members attended workshops and training sessions. This helped them develop the frontend and backend more effectively.
- Managing Team Members
 - Clear communication channels were established, and roles and responsibilities were clearly defined. Regular team meetings were held to track progress and address any challenges faced by members.
- Tight Deadlines
 - The team used ClickUp to manage time and resources efficiently. They followed Agile practices like sprints, breaking the project into smaller, manageable tasks to ensure timely completion.

8.Communication and Reporting

Within the Team

- Short team meetings were held daily to discuss progress, challenges, and upcoming tasks.
- Quick updates and discussions were handled through messaging apps (WhatsApp).
- Important project updates and documents were shared.
- Work progress was tracked using JIRA or ensure deadlines were met.
- Weekly meetings were conducted to discuss project major challenges.

with the Client

- Reports were sent to update the client on the project's status.
- Scheduled Meetings
- The team conducted live demonstrations at key project stages to showcase completed features and gather feedback.

Reporting Mechanisms

- Team members provided updates on completed tasks, ongoing work, and blockers in JIRA.
- Issues and bugs were recorded in JIRA.
- Meetings and presentations for client updates and discussions.

9. Testing and Quality Assurance

Unit Testing:

After finishing each small part of the system, we will test that part alone to make sure it works correctly. This helps catch problems early.

Integration Testing:

When two or more parts of the system are connected, we will test them together to ensure they work well as a team.

System Testing:

Once all parts of the system are combined, we will evaluate the entire system to confirm it works as expected in real-life situations, like managing hotel rooms, reservations, and billing.

User Acceptance Testing (UAT):

Before launching the system, we will ask the client and users to test it. Their feedback will help us ensure the system meets their needs and is ready for use in hotels.

Regression Testing:

Every time we add new features or fix issues, we will re-test the system to make sure the changes didn't break anything that was already working.

Usability Testing:

We will check if the system is easy and pleasant to use for hotel staff and guests. If there are any problems, we will improve the design and functionality.

Quality Assurance:

Along with testing, we will review the code regularly, follow good coding practices, and keep improving how we build the system. This ensures the final product is of the highest quality.

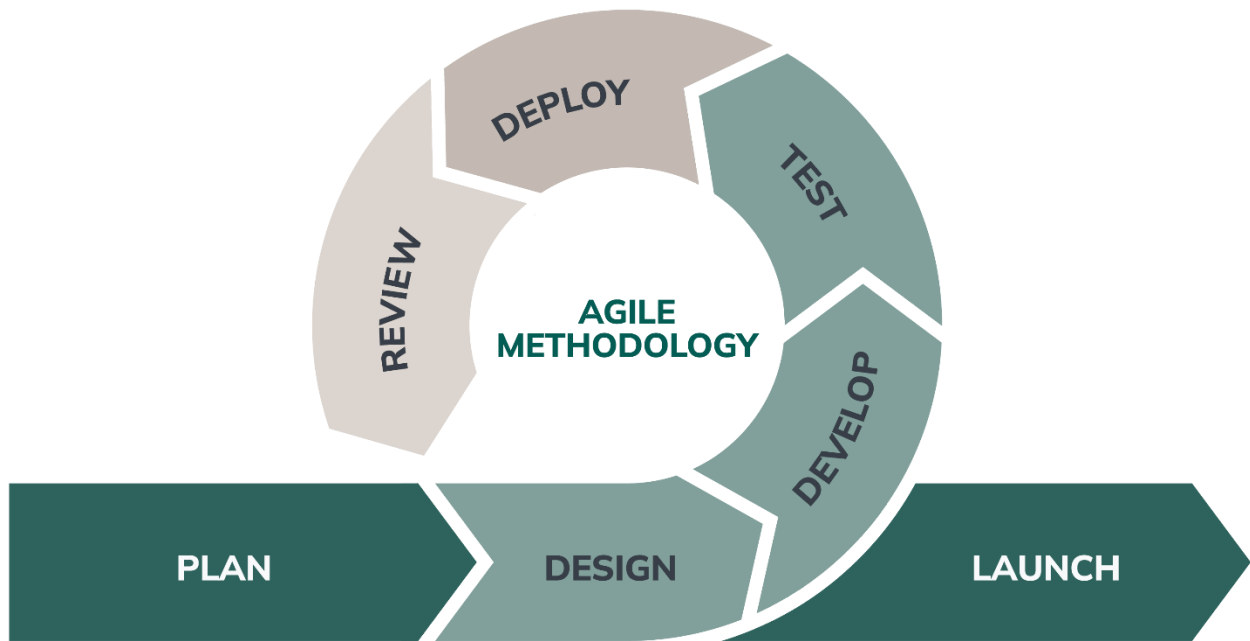


Figure 2: Agile Cycle

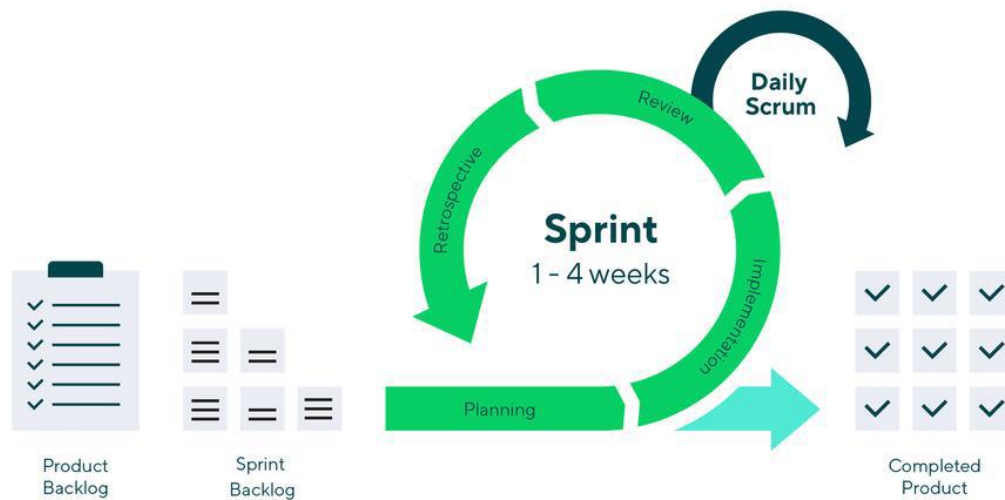


Figure 3: Scrum Cycle

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11.Appendix



Figure 5 : Pool Details



Figure 4: Special Event



Figure 7 : Room Reservation



Figure 6 : Event Booking